

JIVITESH SHARMA

APPLIED AI / MACHINE LEARNING ENGINEER

GENERATIVE AI / AGENTIC SYSTEMS / PRODUCTION AI

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Professional Profile

Applied AI and Machine Learning Engineer with 5+ years building production systems across generative and agentic AI, NLP, graph learning, computer vision, and geospatial intelligence. Work spans modelling, data pipelines, evaluation, product integration, and cloud delivery.

At HERE Technologies, built production graph and geospatial ML for map intelligence: GNN-based road-network alignment, conflation and clustering across heterogeneous sources, a LightGBM lane-attribute classifier at approximately 74% F1, stateful self-healing pipelines, and BMW Urban Cruise Control evaluation work that improved measured KPIs by approximately 5%.

At MSCI, advanced petabyte-scale document extraction from prompting-based LLM workflows into evaluated multi-agent systems on GCP and Vertex AI. Owned retrieval, structured outputs, guardrails, agent evaluation, and human review, moving multiple datasets from below 50% precision and recall to >90% recall and >80% precision.

Bring a systems view across classical ML, deep learning, graph ML, retrieval, agents, evaluation, and production delivery.

Professional Capabilities

Generative & Agentic AI	- Large Language Models (LLMs) - Agentic AI - RAG - Hybrid Search - Semantic Reranking - LangGraph - LangChain - CrewAI	- Model Context Protocol (MCP) - Tool Calling - Multi-Agent Orchestration - Prompt & Context Engineering - Structured Outputs - Guardrails - Human-in-the-Loop - Agent Evaluation & Tracing
Machine Learning & Deep Learning	- Classical ML - Deep Learning - PyTorch - Scikit-learn - LightGBM - Predictive Modelling	- Feature Engineering - Model Evaluation - SHAP - NLP - Computer Vision (CNNs) - LSTMs
Graph & Geospatial AI	- Graph Neural Networks - PyTorch Geometric - NetworkX - GraphSAGE - HinSAGE	- GCNs - GeoPandas - Geospatial Analytics - Network Alignment & Conflation
Cloud & Production AI	- GCP - Vertex AI - AWS - Cloud Computing	- Production AI Delivery - Evaluation Pipelines - CI/CD - APIs
Programming, Data & Developer Tools	- Python - SQL & SQLite - Pandas - NoSQL	- Data Visualization - Cursor - Claude Code - OpenAI Codex

Work Experience / AI & NLP Engineering

April 2024-Present

AI/NLP Engineer at MSCI

Advanced petabyte-scale enterprise document extraction from a prompting-based LLM pipeline into evaluated, production-grade agentic AI.

Enterprise Scope

Worked within MSCI's AI Engineering and GenAI Solutions organization, where cross-functional teams across Data, Research, Engineering, and Product built production-grade AI for regulated enterprise data. My ownership centered on petabyte-scale document extraction across annual reports, company disclosures, PDFs, and proprietary enterprise data sources. I connected LLMs, RAG, agents, tool calling, APIs, and human-in-the-loop review into structured, traceable outputs for downstream data products, treating precision, recall, grounding, and reliability as product KPIs.

End-to-End Ownership

Owned the full path from evidence discovery and data preparation to validated structured outputs. The original pipeline preprocessed documents, ranked candidate pages with BM25, assembled context for an LLM, and post-processed responses into required schemas. I improved page detection with BERT-based semantic reranking and worked across few-shot and methodology-aware prompting, cleansing, validation, hallucination checks, ground-truth construction, and failure analysis, improving retrieval and downstream extraction together.

Measured Improvement

Single-handedly advanced multiple extraction datasets from below 50% precision and recall to >90% recall and >80% precision through an architectural shift from a prompting-based LLM system to an agentic workflow. Built assessment scripts that calculated data-point precision and recall, compared retrieval, prompt, and model variants, exposed failure modes, and supported regression gates for meaningful changes. This created a repeatable route from experimentation to KPI-backed production delivery.

AI Quality Engineering

Systematically debugged real-world AI behavior and improved prompts, retrieval, context, tools, orchestration, guardrails, and model selection. Combined evidence grounding, schema validation, post-processing controls, and unsupported-claim checks with model and agent evaluations covering tool correctness, handoffs, grounding, and end-to-end task completion. Evaluation results became the operating feedback loop for deciding readiness, risk, and the right points for human review.

2024-Present

Production AI Architecture, Evaluation & Delivery

Agentic Redesign

Re-architected extraction from retrieval-ranked LLM calls into a CrewAI-based, production-grade multi-agent application. A prompt-builder agent interpreted methodology documents; extraction agents invoked document tools and APIs; and cleansing and validation agents normalized outputs and challenged unsupported results. Designed sequential and hierarchical flows with tool use, reusable skills, and Agent2Agent (A2A) communication according to task complexity and validation needs.

Reusable AI Platform

Developed MCP servers, agent tools, reusable skills, integrations, evaluation frameworks, and deployment accelerators for document extraction. Built context and memory-service patterns that retained methodologies, retrieved evidence, intermediate decisions, and prior tool results without overwhelming model context windows or weakening traceability. Converted dataset-specific failures into reusable prompt, context, tool, and evaluation components that strengthened later deployments.

Product & Cloud Delivery

Extended the platform into an AG-UI conversational extraction product where users could upload documents, describe the business meaning of requested data, and guide extraction interactively. Deployed agentic workflows on GCP and Vertex AI while preserving retrieved evidence, structured outputs, validation, and human review. The work contributed to MSCI's wider AI-first data-acquisition program supporting high double-digit cost reduction, multi-million-dollar savings, and faster launches.

Cross-Functional Delivery

Worked as the technical bridge across methodology, data, product, research, platform, and engineering teams. Translated domain intent into schemas, acceptance criteria, evaluation datasets, and executable delivery plans; adapted solutions inside existing pipelines and enterprise constraints; and aligned contributors around precision, recall, grounding, and workflow-adoption goals. Used traceable evidence and failure analysis to explain tradeoffs, unblock delivery, and return field learning to the shared platform.

2022-March 2024

Data Scientist at HERE Technologies

Recognized as a critical employee for contributions to production mapping and ML systems.

- Developed and tuned a LightGBM model to classify lane-marking style and color. Engineered predictive features, used SHAP importance to select high-value inputs, and achieved approximately 74% F1.
- Implemented a GNN-based alignment approach for matching road-network graphs from two sources. Generated node embeddings that captured positional, attribute, and higher-order neighborhood information, then scored candidate matches with cosine similarity.
- Built association and clustering logic for source conflation, improving the identification of equivalent features and the merging of attributes across heterogeneous map sources.
- Implemented a stateful, self-healing map pipeline that continuously incorporated fresher streaming data while preserving conflated feature state. Contributed to a related patent filing that remained under review.
- Built a full-scale evaluation pipeline for BMW Urban Cruise Control map features, identifying false positives and false negatives and guiding production POCs that improved measured KPIs by approximately 5%.
- Developed feature-selection and classification models for UCC attributes and introduced graph-based dimensionality-reduction techniques to reduce false positives and improve product accuracy.

Earlier Professional Experience

Fall/Winter 2021

Data Science Intern at HERE Technologies

Geospatial machine learning and graph-based network alignment.

- Processed and analyzed geospatial data with GeoPandas, Shapely, PyProj, and GeoJSON.
- Transformed map tiles across coordinate reference systems and supported ML components within an end-to-end mapping and geolocation platform.
- Worked in Agile delivery workflows and used CI/CD practices to maintain traceable, current software versions.
- Researched graph-learning approaches for network alignment using NetworkX, StellarGraph, PyTorch Geometric, GraphSAGE, HinSAGE, and GCNs.

Summer 2021

Consultant Data Scientist at Edelman

Digital Advisory team focused on NLP, market intelligence, and social listening.

- Built ML and NLP pipelines for multiple client projects on Edelman's Digital Advisory team.
- Delivered sentiment and text-analysis work, including market-strategy research for Qualcomm and web-scraping/NLP-based gap analysis for Logitech.
- Produced biweekly and monthly reports for multinational clients using Talkwalker and other social-listening data, translating consumer signals into actionable insights.

Winter 2020

Conversational Nutrition Assistant using Rasa

- Built a nutrition assistant in Rasa, defining intents, entities, conversation stories, and custom actions.
- Integrated the Edamam and Spoonacular APIs to generate meal plans, calculate BMI, and return health information.

Fall 2020

IBM Capstone: COVID-19 Time-Series Analysis

- Developed an end-to-end analysis notebook covering ETL, EDA, visualization, feature engineering, and model training.
- Built reusable scikit-learn pipelines for data transformation and used Matplotlib to explore temporal patterns.
- Created windowed sequences for an LSTM and compared its performance with SVM and dense neural network baselines.

Fall 2020

Rock-Paper-Scissors CNN Classifier

- Worked with a team to create and label an original hand-image dataset for rock, paper, and scissors classification.
- Prepared train, validation, and test splits; resized and normalized image tensors; and trained CNNs with different filters, depths, padding, and layers.
- Selected the final model by comparing validation loss and accuracy across experiments.

Summer 2020

Wine Classification using H2O.ai

- Explored a Kaggle wine dataset with Matplotlib and Seaborn and used PCA to identify variables that best separated wine classes.
- Trained random forest and gradient boosting classifiers in H2O.ai and compared their performance.

Summer 2019

2019 General Assembly Election Analysis

- Analyzed historical election results and visualized the leading parties using Matplotlib and Seaborn.
- Scraped public tweets from Narendra Modi's account and used TextBlob sentiment analysis to assess social-media signals related to the BJP.
- Built a linear regression model to estimate the BJP's vote count in the 2019 election.

Sep-Dec 2018

SP Jain School of Global Management Projects

- Neural Network Architecture: Researched the history, fundamentals, and major architectural families of neural networks.
- Amazon Alexa vs. Google Home: Compared assistant capabilities and explored opportunities for adaptive, self-learning products.
- Intel Movidius and Google AIY: Prototyped embedded computer-vision and voice-recognition applications.
- Alexa and Raspberry Pi Security Camera: Built a facial-recognition security camera prototype using a Raspberry Pi camera, Alexa integrations, Kairos, and AWS services.

Earlier Experience, Certifications & Education

2016-2017

Early Technology & Leadership Experience

- Software Development Intern, Appinventiv - supported early-stage application development, collaborated with software teams, and learned product delivery practices.
- TEDx Event Organizer
- TCS IT Wiz Qualifier

Completed Certifications

DeepLearning.AI Deep Learning Specialization

- Neural Networks and Deep Learning
- Improving Deep Neural Networks
- Structuring Machine Learning Projects
- Convolutional Neural Networks
- Sequence Models

IBM Advanced Data Science Specialization

- Fundamentals of Scalable Data Science
- Advanced Machine Learning and Signal Processing
- Applied AI with Deep Learning
- Advanced Data Science Capstone

Education

2018-2021

Bachelor of Data Science

SP Jain School of Global Management, Sydney

2016-2018

IB Diploma (International Baccalaureate)

Pathways World School Aravali, Gurgaon | 36 points